Task 3: Neural LMs + Evaluation (10 points)

1. Neural Language Models: components

1. **Embedding layer** – maps tokens to dense vectors
2. **Encoder** (RNN, CNN, Transformer) – computes contextual representations
3. **Output softmax** – converts hidden states into a probability distribution over the vocabulary

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2. Recurrent models

Why RNN/LSTM/GRU work well:

- They process sequences sequentially
- They maintain a hidden state that summarizes past context

Hidden state stores:

- Syntactic structure
- Semantic context
- Long-range dependencies (especially in LSTM/GRU)

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3. Convolutional models

A **1D CNN** slides convolution filters over token embeddings.

What it captures well:

- Local patterns (phrases, collocations)
- Short-range dependencies
- Parallel computation (faster than RNNs)

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4. Evaluation: Perplexity

Perplexity (PPL):

PPL=exp(cross-entropy)\text{PPL} = \exp(\text{cross-entropy})PPL=exp(cross-entropy)

Lower perplexity means the model assigns higher probability to the true sequence.
It measures how “surprised” the model is by the data.

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5. Weight tying

Weight tying means using the same matrix for:

- Input embeddings
- Output softmax weights

Why it helps:

1. Reduces number of parameters
2. Improves generalization
3. Enforces consistency between input and output word representations

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6. Analysis & interpretability methods

Two valid methods:

- **Gradient-based saliency / attribution**
- **Probing classifiers** (test what linguistic information is encoded)

Other acceptable answers:

- Neuron activation analysis
- Attention/token importance

- Error bucket analysis